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Gradient Embedding and Modal Complementarity Networks for Audio-Video Depression Level Prediction

Mingyue Niu, Zhongping Zhang, Chang Gao, Yan Zeng, Ning Fan, and Shiqing Zhang

***Abstract*—Physiological studies show distinct audio-video behaviors correlate with depression levels, leading some researchers to use Transformer architecture to predict depression levels. The positional embedding in Transformer assigns a positional marker to each element, but the multi-head attention mechanism operates at the token level and is difficult to examine the temporal changes of elements. This brings limitations in capturing dynamic differences reflecting depression. Besides, previous works adopt element-wise operation, fully connected layers or attention mechanisms for modal complementarity. However, they don't specify what the supplemental representation is, and don't explore to what extent the supplemental representation is accepted. This hinders the model to characterize depression cues. Therefore, we propose the Gradient Embedding (GE) and the Modal Complementarity (MC) modules. The GE module embeds the gradient into sequences to perceive the dynamics of elements, thereby helping to capture depression-related temporal patterns. Besides, we stipulate the supplementary representation as the difference between two modal representations. In this way, the MC module quantifies to what extent the supplemental representation is accepted by a weighted combination of unimodal representation and supplemental representation, thereby achieving effective modality complementarity. Experimental results (RMSE=6.42/6.23, MSE=5.06/5.00) on AVEC 2013 and AVEC 2014 datasets illustrate the progressiveness of our method.**

***Index Terms*—Depression level prediction, Gradient embedding, Modal complementarity, Audio-Video.**

I. Introduction

In recent years, due to the increase of unstable factors, the difficulties and pressures people face have also increased. The persistent stress in real life leads to mental health problems and loss of interest in life, making people more susceptible to depression. According to data released by the World Health Organization in 2023 [1], the number of global depression patients has reached nearly 300 million. However, the uneven distribution and scarcity of mental health resources have posed obstacles to the diagnosis of patients and delayed optimal treatment time, resulting in irreversible consequences. Therefore, researchers hope to use artificial intelligence technologies to help people understand their mental health status in a timely manner and reduce the occurrence of unfortunate events.

TABLE I
Correspondence between BDI-II score range and depression degrees.

BDI-II score range	Depression Degrees
0–13	None
14–19	Mild
20–28	Moderate
29–63	Severe

Coincidentally, physiological and psychological studies [2], [3] have found differences in speech signals and facial activities between depressed and healthy individuals. Those facts indicate that it is feasible to use artificial intelligence technology to analyze individual speech and facial data to predict the severity of depression. Motivated by those studies, considering the powerful modeling capabilities of Transformer architecture, various algorithms [4]–[11] are developed to extract the representations characterizing depression cues from audio and video modalities for predicting BDI-II scores, which can be correlated with the degree of depression through Table I. For those works, they mainly consist of **two stages**: (1) the extraction of unimodal depression cue representations; (2) the fusion of multimodal representations. Although some progress has been made in previous methods, there are still some issues to be studied with regard to those two stages.

For the stage (1), previous works often feed audio or video sequences into the Transformer encoder to extract depression cue representation. The Position Embedding (PE) step in the Transformer architecture assigns a fixed positional marker to each element[1], but the Multi-Head Self-Attention (MHSA) mechanism is a operation on the token level and is difficult to examine the temporal changes of each element, which is not conducive to capturing dynamic patterns related to depression in the long-range modeling process. Besides, although some works [4], [5], [10], [12] combine convolutional layers and attention mechanisms, they don't embed local changes into the input sequence. In other words, those works only examine

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[1] In this paper, an element refers to a certain position in the input sequence, that is, assuming the input sequence is $\boldsymbol{X} \in \mathbb{R}^{M\times N}$, then the i-th ($i = 1, \cdots, M$) row and j-th ($j = 1, \cdots, N$) column of $\boldsymbol{X}$ are an element, which can be referred to as $\boldsymbol{X}_{ij}$.

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Δx_i $(i = 1, 2, \cdots, N,$ where N is the number of elements$)$ without jointly examining x_i and Δx_i, which reduces the model's ability to extract discriminative information related to depression. **For the stage (2)**, previous works [4], [6], [10], [13] often use element-wise operations, Fully Connected (FC) layers or attention mechanisms to achieve modal complementarity. However, all of those manners don't specify what the supplementary representation is, nor do they explore the acceptance degree of the modality towards the supplementary representation, which is difficult to achieve effective modal complementarity and poses an obstacle to improving the model's ability to characterize depression cues.

To alleviate the above problems, we propose two novel modules, namely the Gradient Embedding (GE) module and Modal Complementarity (MC) module. For the GE module, it uses gradient templates to extract multiply order gradient of elements and embeds them into the input sequence through convolutional layers. Note that, in this paper, the gradient refers to the gradient of each element in the sequence along the time direction, that is, the change of each element along the time direction, rather than the gradient in backpropagation. This can help the model perceive the temporal changes of elements while modeling long-range dependencies, thereby promoting MHSA to capture the dynamic differences presented by individuals with different levels of depression in the audio and video modalities. For the MC module, it obtains the supplementary representation of the audio (video) modality to the video (audio) modality based on the stipulation, which is given by the difference between modal representations in this paper. Furthermore, we use the gating mechanism to handle the concatenation of video (audio) modality and supplemental representations to obtain the combination weight for determining the acceptance degree of the video (audio) modality for the supplemental representation, thereby achieving effective modality complementarity and enhancing the model's ability to characterize depression cues. In this way, we integrate the GE and MC modules into a framework called GEMC network to predict depression levels. The experimental results on AVEC 2013 [14] and AVEC 2014 [15] datasets show that our method is progressiveness.

In general, the main contributions of this paper can be summarized as follows:

(1) Our proposed GE module embeds the gradient information into the input sequence to enable MHSA to perceive temporal changes of elements while long-range modeling, which helps capture depression cues hidden in speech and facial dynamics.

(2) In this paper, we provide a stipulation of supplementary representation. Based on this stipulation, our proposed MC module can not only explicitly obtain supplementary representations of modalities, but also automatically determine the degree to which the supplementary representations are accepted. Therefore, the MC module achieves effective modal complementarity and enhances the ability to characterize depression cues.

(3) We verify the rationality of the stipulation of supplementary representation through experiments. At the same time, we also demonstrate the progressiveness of our constructed GEMC architecture through experiments on AVEC 2013 and AVEC 2014 databases.

Below, our paper is organized as follows: In the second section, the relevant works are briefly given; In the third section, the details of our method are described; In the fourth section, relevant experimental results and analysis are presented; In the fifth section, our work is summarized.

II. Related Works

As mentioned earlier, the extraction of unimodal depression cue representations and the fusion of multimodal depression cue representations are the two main stages for the depression level prediction task based on audio and video. Therefore, in this section, we will briefly review some previous works about those two stages.

A. Depression Cue Representation Extraction from Audio Modality

For the audio modality, Liu et al. [16] proposed an Attention Guided Bi-direction Temporal-aware Network (AGBiTNet), which not only examined the semantic content of Mel Frequency Cepstral Coefficients (MFCC), but also used the FC layers to generate the attention weight for emphasizing useful spectral regions. Moreover, they combined Huber and Generalized End-to-End (GE2E) losses to predict an individual's depression level. Considering the advantages of Transformer architecture in modeling long-range dependencies, Flores et al. [6] used a pre-trained model VGGish to obtain the token sequence of speech, which was processed by BiLSTM and fed into a Transformer encoder to obtain the depression cue representation of the audio modality. From a time-frequency perspective, Chen et al. [17] constructed a Temporal Frequency Features Fusion Network (TTFNet). They first used quaternion representation to encode the Mel spectrum and its derivatives, and adopted quaternion VisionLSTM to examine the synergistic relationship between quaternions. Next, they used the wav2vec 2.0 model to investigate the temporal changes of speech waves. Finally, the XConformer module composed by MHSA and convolution layers was to implement the interaction between two sequences to assess the depression severity of individuals. In the work of [18], the author used auto-masked techniques to enhance context learning and examined the long-range dependencies of speech spectrogram through Transformer encoders. Finally, they highlighted the depression-related content in the spectrogram from a time-frequency perspective.

B. Depression Cue Representation Extraction from Video Modality

For the video modality, in the work of [19], Hu et al. designed the Landmark Image Landmark (LIL) and Global Vision Transformer (CGViT) modules. The LIL module was used to extract facial features and landmarks, and generated attention weights through matrix multiplication to achieve interaction between the two types of features. The GCViT module was used to embed global information into visual

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representations to enhance the ability to model long-term facial dynamics. The authors in [12] developed a Dual Attention and Element Recalibration (DAER) network, which was mainly composed of Dual Attention (DA) and Element Recalibration (ER) blocks. The DA block adopted self-attention and bilinear-attention to examine the facial changes and the impact of feature components on the target task. The ER block was to embed the global information into the tensor for improving the ability of characterizing depression cues. Inspired by the good experimental performance of Transformer architecture in the field of image processing, in the work of [6], the author used the OpenFace toolbox to extract facial features and BiLSTM to capture the temporal relationship for obtaining the sequence. Finally, they used the Transformer encoder to generate the representation of depression cues. Similarly, Tao et al. [7] viewed facial keypoint sequences as token sequences and used attention mechanisms to emphasize useful content from a spatiotemporal perspective. Then, the Transformer encoder was used to capture long-range dependencies to extract information related to depression. The authors in [8] employed the Transformer encoder and convolutional layers to extract the global and local features, which were combined to predict the BDI-II score of an individual.

For the above works, although some works use the Transformer architecture to extract depression cues, they still adopt PE to make the model perceive the relative order of tokens. This makes it difficult for the model to balance the temporal changes of elements while modeling long-range dependencies, which is not conducive to capturing the speech and facial dynamics presented by individuals with different depression levels.

C. Depression Level Prediction by Fusing Audio and Video Modalities

In order to achieve complementary advantages of audio and video modalities, many researchers have proposed various fusion algorithms to improve the accuracy of predicting depression levels. In those methods, element-wise operation, FC layers or attention mechanisms are three common ways to achieve complementary. Uddin et al. [9] obtained spatiotemporal representations of audio and video by combining CNN with BiLSTM. Then, the audio and video features were fused with Multimodal Factorized Bilinear (MFB) pooling, which achieved the complementarity of different features through element-wise multiplication operation. Zhang et al. [20] designed Feature Alignment and Aggregation (FAA) and Sequence Encoding and Aggregation (SEA) modules. The FAA module was to evaluate the relevance of audio and video modalities for element-wise addition.

Different from them, the work of [11] proposed a Parallel Multiscale Bridge Fusion Network (PMBFN). This model cascaded CNN and LSTM layers to extract long-term spatiotemporal representations of audio and video modalities, and then aggregated one modality with another to achieve interaction between modalities, and completed modality complementarity through FC layers. Likewise, Flores [6] used Transformer to obtain audio and video representations, and fused the

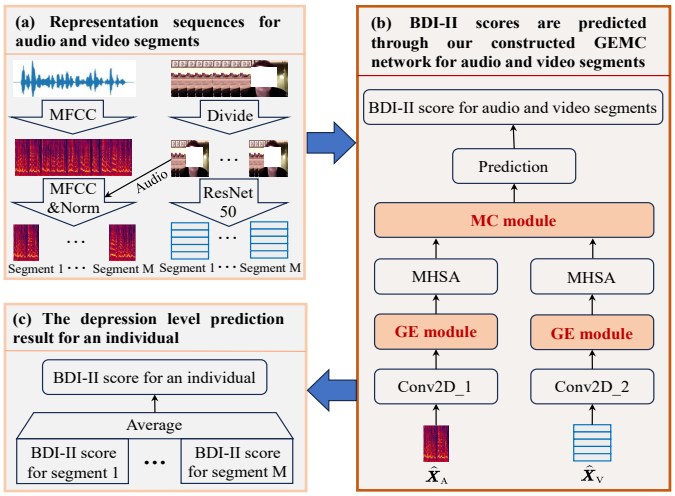

Fig. 1. The complete flowchart is provided using our constructed GEMC network to predict an individual's depression level through audio and video modalities. "ResNet50" is used to encode facial images and obtain the representation sequence for the video. In this prediction process, there are three steps: (a) The long-term audio and video are divided into segments with fixed duration, and gain the representation sequences for audio and video segments. (b) The representation sequences for the segments of two modalities are input into our constructed GEMC network to predict the BDI-II score for the segment. (c) The average of scores of all segments is regarded as the depression level for the corresponding individual. "GE" and "MC" are two our proposed modules namely Gradient Embedding and Modal Complementarity modules. "M" is the number of segments obtained by dividing the long-term audio and video. "$\widehat{\boldsymbol{X}}_{\mathrm{A}}$" and "$\widehat{\boldsymbol{X}}_{\mathrm{V}}$" are denoted as the representations for audio and video segments.

two modalities through a FC layer to predict an individual's depression level. For combining different two ways, Zhou et al. [13] jointed global and local information to characterize the temporal behaviors, and achieved early interaction between different modalities through element-wise addition operations. Furthermore, they used mixed features as weights to add audio and video representations to achieve complementary advantages and obtain depression level prediction results.

From the previous methods, they use element-wise operations, FC layers or attention mechanisms to achieve modal complementarity. However, they lack a clear stipulation about the supplementary representation and have not explored to what extent the supplementary representation is accepted. This makes it difficult for them to achieve effective and sufficient modal complementarity, thereby affecting the model's ability to characterize depression cues.

III. Our Gradient Embedding and Modal Complementarity Networks for Depression Level Prediction

In this paper, a GEMC network is constructed to predict an individual's depression level through audio and video modalities. Specifically, we divide the long-term audio and video into short-term segments with fixed duration, and obtain representation sequences of segments with Mel-filters and pre-trained ResNet50. Next, the representation sequences of two modalities are fed into the GEMC network to obtain the BDI-II scores of the audio and video segments. Finally, the average of those scores is considered as the individual's BDI-II score. Fig. 1 provides a complete flowchart for predicting depression

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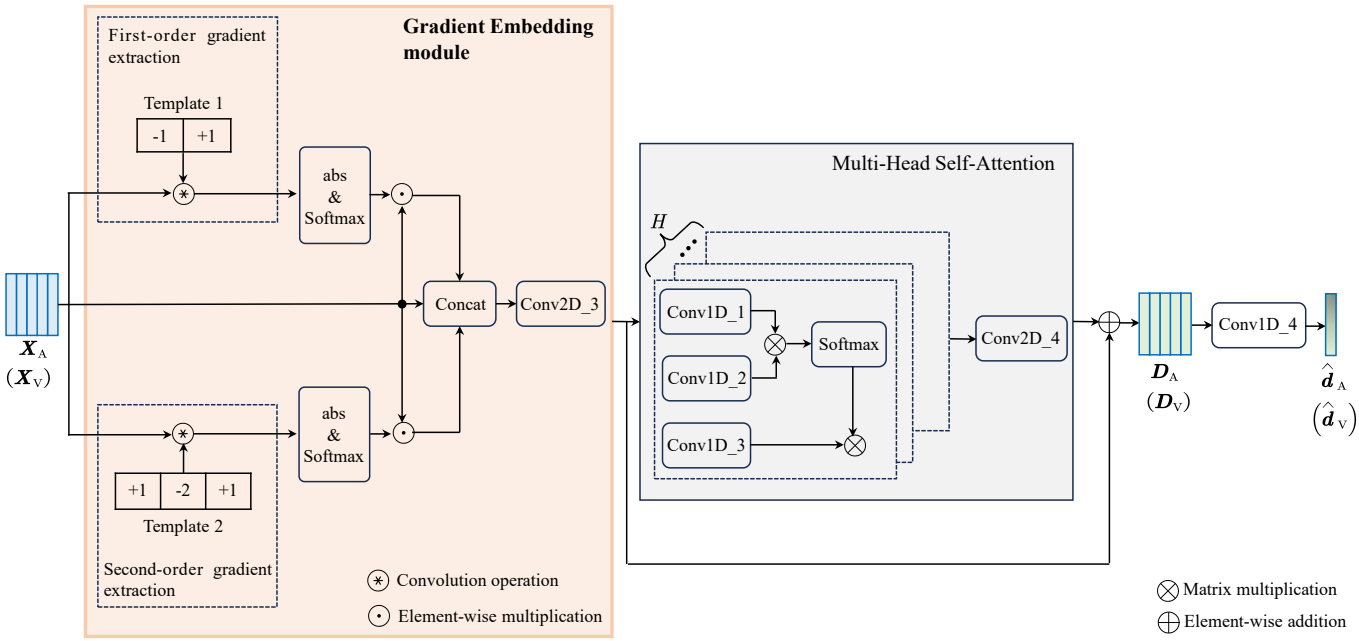

Fig. 2. The combination of our proposed GE module and Multi-Head Self-Attention (MHSA) mechanism. There is a Layer-Normalization layer after $\oplus$ operation, but for simplicity, we do not draw it in this figure. H refers to the number of heads in the MHSA mechanism. abs is the absolute value function. Concat is short for concatenation. Note that the convolutional layers have padding operations.

levels using the GEMC network. As shown, "GE" and "MC" are two core modules. Thus, in the following section, we will elaborate on the details of those two modules.

A. Representation Sequences Generation for Our Network

As shown in Fig. 1, we encode the audio and video modalities into representation sequences and input them into our constructed GEMC network. Therefore, in this section, the process of representation sequence generation for two modalities is elaborated.

For the audio modality, we use the Librosa toolbox to generate the MFCC of a long-term speech. The maximum and minimum values of each frequency band are denoted as $f_{\max}$ and $f_{\min}$, respectively. In this way, the pre-set window length and repetition rate are used to divide a long video into video segments. The audio in the video segment is then extracted and converted into the corresponding MFCC segment, which is normalized band-by-band using $f_{\max}$ and $f_{\min}$. For convenience, in this paper, we denote each MFCC segment as $\widehat{\boldsymbol{X}}_{\mathrm{A}} \in \mathbb{R}^{P\times\widehat{N}}$, where P and $\widehat{N}$ are the number of frequency bands and frames, respectively. Furthermore, for computational efficiency, we use the "Conv2D_1" layer to merge adjacent frames of $\widehat{\boldsymbol{X}}_{\mathrm{A}}$. The corresponding result is denoted as $\boldsymbol{X}_{\mathrm{A}} \in \mathbb{R}^{P\times N}$, where $N < \widehat{N}$.

For the video modality, we divide a long-term video into segments with fixed frames and extract facial images for each frame using the OpenFace toolbox. Then, each frame in the segment is fed into the trained ResNet50 model with ImageNet, and the output of the "avg_pool" layer is regarded as the encoding vector for the facial image. In this way, a video segment is encoded into a representation sequence denoted as $\widehat{\boldsymbol{X}}_{\mathrm{V}} \in \mathbb{R}^{\widehat{Q}\times M}$, where $\widehat{Q}$ and M are the dimension of encoding vector and the number of frames, respectively. Note that the depression level prediction is a regression task. Thus, like [21]–[23], we use the ImageNet dataset to train ResNet50 with Euclidean loss and ReLU instead of cross-entropy loss and Softmax. Then, the trained parameters of ResNet50 are frozen and used as an encoder to obtain the vector encoding results of video frames. Similar to the audio modality, we use "Conv2D_2" layer to reduce the dimension of $\widehat{\boldsymbol{X}}_{\mathrm{V}}$ and record the result as $\boldsymbol{X}_{\mathrm{V}} \in \mathbb{R}^{Q\times M}$, where $Q < \widehat{Q}$.

B. Gradient Embedding Module for Facilitating Depression Cues Capture

As physiological and psychological studies [24], [25] shown, the differences among individuals with different depression levels are mainly reflected in the dynamic changes of speech and facial activities. And $\boldsymbol{X}_{\mathrm{A}}$ and $\boldsymbol{X}_{\mathrm{V}}$ happen to be temporal sequences. Thus, when using Transformer architecture to process $\boldsymbol{X}_{\mathrm{A}}$ and $\boldsymbol{X}_{\mathrm{V}}$, dynamic information embedding is more helpful in capturing depression cues than positional information embedding. Based on this consideration, we propose the GE module, which embeds gradient information into each element of the sequence to facilitate MHSA's capture of depression cues. Fig. 2 gives the pipline of the combination of our proposed GE module and MHSA.

Specifically, we use "Template 1" and "Template 2" in Fig. 2 to perform convolution operations with $\boldsymbol{X}_{\mathrm{A}}$ ($\boldsymbol{X}_{\mathrm{V}}$) for calculating the first- and second-order gradient of each element. The corresponding results are denoted as $\widehat{\boldsymbol{W}}_{\mathrm{A1}} \in \mathbb{R}^{P\times N}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}} \in \mathbb{R}^{P\times N}$ ($\widehat{\boldsymbol{W}}_{\mathrm{V1}} \in \mathbb{R}^{Q\times M}$ and $\widehat{\boldsymbol{W}}_{\mathrm{V2}} \in \mathbb{R}^{Q\times M}$). It is worth noting that the values of $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ at each position can be approximated as the first-order and second-order derivatives of the corresponding position in $\boldsymbol{X}_{\mathrm{A}}$. That is to say, $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ give the changes in each position of $\boldsymbol{X}_{\mathrm{A}}$ and the speed of changes, which is helpful for capturing differences among individuals with different depression levels.

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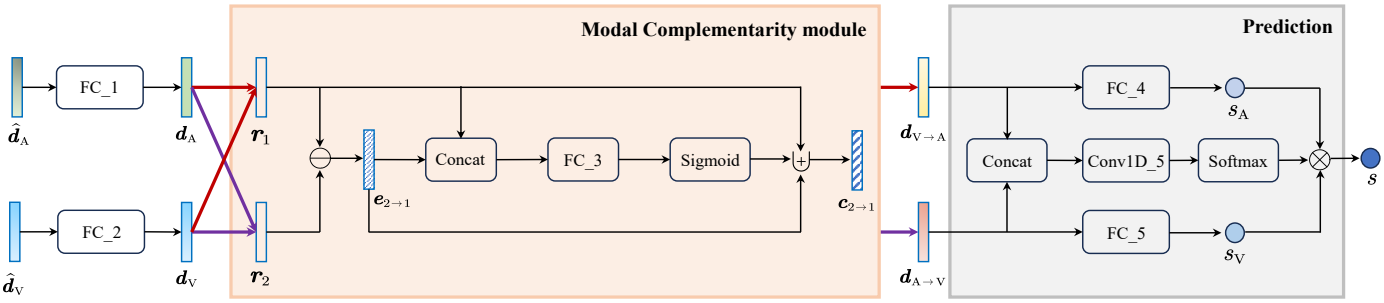

Fig. 3. Our proposed MC module and the prediction process. $\ominus$ specifically refers to $\boldsymbol{r}_2 - \boldsymbol{r}_1$. $\uplus$ is defined as Eq. (8). Concat is short for concatenation.

Furthermore, in order to highlight the magnitude of the change rather than its direction, we perform taking absolute value operations on $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ ($\widehat{\boldsymbol{W}}_{\mathrm{V1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{V2}}$). For the sake of simplicity, the corresponding results are still referred to as $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ ($\widehat{\boldsymbol{W}}_{\mathrm{V1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{V2}}$).

After the above process, as shown in Eq. (1), we embed $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ ($\widehat{\boldsymbol{W}}_{\mathrm{V1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{V2}}$) into $\boldsymbol{X}_{\mathrm{A}}$ ($\boldsymbol{X}_{\mathrm{V}}$) through the attention mechanism. The corresponding results are taken as $\widehat{\boldsymbol{X}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{X}}_{\mathrm{A2}}$. What one can observe is that since $\widehat{\boldsymbol{W}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{W}}_{\mathrm{A2}}$ are two matrices with all positive numbers, whether Eq. (1) excites or suppresses the elements of $\widehat{\boldsymbol{X}}_{\mathrm{A}}$ does not change the physical properties of the sequence, which is advantageous for capturing depression cues. In this way, similar to PE in Transformer, we add $\widehat{\boldsymbol{X}}_{\mathrm{A1}}$ and $\widehat{\boldsymbol{X}}_{\mathrm{A2}}$ to the input sequence. This process can be achieved through Eq. (2) and the corresponding result is denoted as $\widehat{\boldsymbol{D}}_{\mathrm{A}}$ ($\widehat{\boldsymbol{D}}_{\mathrm{V}}$).

$$\begin{cases} \widehat{\boldsymbol{X}}_{\mathrm{A1}} = \boldsymbol{X}_{\mathrm{A}} \odot \mathrm{Softmax}\left(\widehat{\boldsymbol{W}}_{\mathrm{A1}}\right) \in \mathbb{R}^{P\times N} \\ \widehat{\boldsymbol{X}}_{\mathrm{A2}} = \boldsymbol{X}_{\mathrm{A}} \odot \mathrm{Softmax}\left(\widehat{\boldsymbol{W}}_{\mathrm{A2}}\right) \in \mathbb{R}^{P\times N} \end{cases}, \tag{1}$$

where $\mathrm{Softmax}(\cdot)$ is operated on every row of input matrix.

$$\widehat{\boldsymbol{D}}_{\mathrm{A}} = \mathcal{C}_2\left(\mathrm{Concat}\left(\boldsymbol{X}_{\mathrm{A}}, \widehat{\boldsymbol{X}}_{\mathrm{A1}}, \widehat{\boldsymbol{X}}_{\mathrm{A2}}\right)\right) \in \mathbb{R}^{P\times N}, \tag{2}$$

where $\mathcal{C}_2(\cdot)$ refers to a Conv2D layer. $\mathrm{Concat}(\cdot,\cdot,\cdot)$ means concatenating three matrices along the channel axis.

It is necessary to point out that the sequence properties of $\widehat{\boldsymbol{D}}_{\mathrm{A}}$ ($\widehat{\boldsymbol{D}}_{\mathrm{V}}$) have not been changed through the above process. In this way, we adopt the MHSA mechanism to model long-range dependencies and extract the discriminative information among individuals with different depression levels as shown in Eq. (3). Finally, for the stability of training, the residual connection is adopted. The corresponding result is still recorded as $\boldsymbol{D}_{\mathrm{A}} \in \mathbb{R}^{P\times N}$ ($\boldsymbol{D}_{\mathrm{V}} \in \mathbb{R}^{Q\times M}$).

$$\boldsymbol{D}_{\mathrm{A}} = \mathcal{C}_2\left(\mathrm{Concat}\left(\mathcal{A}_1\left(\widehat{\boldsymbol{D}}_{\mathrm{A}}^{\top}\right), \cdots, \mathcal{A}_H\left(\widehat{\boldsymbol{D}}_{\mathrm{A}}^{\top}\right)\right)\right) \in \mathbb{R}^{P\times N}, \tag{3}$$

where $\mathcal{A}_i\left(\widehat{\boldsymbol{D}}_{\mathrm{A}}^{\top}\right)$ $(i = 1, \cdots, H)$ is defined as Eq. (4).

$$\mathcal{A}_i\left(\widehat{\boldsymbol{D}}_{\mathrm{A}}^{\top}\right) = \left(\mathrm{Softmax}\left(\frac{\boldsymbol{K} \otimes \boldsymbol{Q}^{\top}}{\sqrt{P}}\right) \otimes \boldsymbol{V}\right)^{\top} \in \mathbb{R}^{P\times N}, \tag{4}$$

where $\boldsymbol{K} \in \mathbb{R}^{N\times P}$, $\boldsymbol{Q} \in \mathbb{R}^{N\times P}$ and $\boldsymbol{V} \in \mathbb{R}^{N\times P}$ are generated by inputting $\widehat{\boldsymbol{D}}_{\mathrm{A}}^{\top}$ into three Conv1D layers.

Finally, Eq. (5) is used to aggregate the tokens of $\boldsymbol{D}_{\mathrm{A}}$ ($\boldsymbol{D}_{\mathrm{V}}$) to obtain the depression cue representation for the audio (video) modality, denoted as $\widehat{\boldsymbol{d}}_{\mathrm{A}}$ ($\widehat{\boldsymbol{d}}_{\mathrm{V}}$).

$$\widehat{\boldsymbol{d}}_{\mathrm{A}} = \mathcal{C}_1\left(\boldsymbol{D}_{\mathrm{A}}\right) \in \mathbb{R}^{P\times 1}, \tag{5}$$

where $\mathcal{C}_1(\cdot)$ refers to a Conv1D layer.

From the above description, we can see that the proposed GE module embeds gradient information into the input sequence to enable MHSA to perceive the local dynamics of elements while modeling long-range dependencies, thereby facilitating the capture of dynamic differences existing in audio and video modalities. It should be pointed out that although GE is developed by analogy with PE, there are two differences between them: (1) PE is designed to enhance the perception of token order during long-range modeling, while GE is designed to enhance the perception of local changes of elements while long-range modeling; (2) The embedding matrix of PE for all input sequences is the same, while the embedding matrix generated by GE can dynamically change with the input sequence.

C. Modal Complementarity Module for The Fusion of Depression Cue Representations From Audio and Video

As mentioned earlier, we extract depression cue representations from audio and video modalities through combining GE module with MHSA. Hence, it is natural to consider how to achieve complementarity between those two modalities to enhance the model's ability to capture differences among individuals with different depression levels. To this end, as shown in Fig. 3, we propose the MC module.

In order to better achieve modal complementarity, it is necessary to clarify what the supplementary representation between modalities is. To solve this problem, in our paper, we stipulate the supplementary representation as shown in the box below. For this ***Stipulation***, it is very trivial to see that adding $\boldsymbol{r}_1$ to the supplementary representation of $\boldsymbol{r}_2$ to $\boldsymbol{r}_1$ yields $\boldsymbol{r}_2$ i.e., $\boldsymbol{r}_1 + (\boldsymbol{r}_2 - \boldsymbol{r}_1) = \boldsymbol{r}_2$. In other words, the part that $\boldsymbol{r}_1$ lacks relative to $\boldsymbol{r}_2$ or that $\boldsymbol{r}_1$ is different from $\boldsymbol{r}_2$ is the supplement of $\boldsymbol{r}_2$ to $\boldsymbol{r}_1$. Meanwhile, this ***Stipulation*** is similar to the concept of complement sets in set theory. Besides, some researchers [26] have suggested through theoretical deduction that the complementary information between modalities can be obtained from the differences between modalities, and the differences between modalities can serve as complementary

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information. Therefore, the ***Stipulation*** of supplementary representation we provide is reasonable.

> ***Stipulation***: For two modalities m_1 and m_2, if $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ are representations extracted from m_1 and m_2 respectively, then $\boldsymbol{r}_2 - \boldsymbol{r}_1$ is the **supplementary representation** of m_2 to m_1 or the **supplementary representation** of $\boldsymbol{r}_2$ to $\boldsymbol{r}_1$.

From the above ***Stipulation***, it is not difficult to understand that representations extracted from two modalities should have the same dimension. Thus, as shown in Eq. (6), we use two linear layers to make $\widehat{\boldsymbol{d}}_{\mathrm{A}}$ and $\widehat{\boldsymbol{d}}_{\mathrm{V}}$ have the same dimension. The corresponding results are denoted as $\boldsymbol{d}_{\mathrm{A}}$ and $\boldsymbol{d}_{\mathrm{V}}$. In this way, according to the ***Stipulation***, the supplementary representation of the video (audio) modality to the audio (video) modality can be obtained. In the following, for the sake of conciseness, we use $\boldsymbol{r}_1 \in \mathbb{R}^{D\times 1}$ and $\boldsymbol{r}_2 \in \mathbb{R}^{D\times 1}$ to refer to the representations of two modalities to illustrate how our proposed MC module achieves modal complementarity.

$$\begin{cases} \boldsymbol{d}_{\mathrm{A}} = \boldsymbol{W}_{\mathrm{A}} \otimes \widehat{\boldsymbol{d}}_{\mathrm{A}} \in \mathbb{R}^{D\times 1} \\ \boldsymbol{d}_{\mathrm{V}} = \boldsymbol{W}_{\mathrm{V}} \otimes \widehat{\boldsymbol{d}}_{\mathrm{V}} \in \mathbb{R}^{D\times 1} \end{cases}, \tag{6}$$

where $\boldsymbol{W}_{\mathrm{A}} \in \mathbb{R}^{D\times P}$ and $\boldsymbol{W}_{\mathrm{V}} \in \mathbb{R}^{D\times Q}$ refer to two linear layers.

Let $\boldsymbol{e}_{2\to 1} = \boldsymbol{r}_2 - \boldsymbol{r}_1$ be the supplementary representation of $\boldsymbol{r}_2$ to $\boldsymbol{r}_1$. However, the degree to which $\boldsymbol{e}_{2\to 1}$ is accepted by $\boldsymbol{r}_1$ needs to be considered. To this end, as shown in Eq. (7), we concatenate $\boldsymbol{e}_{2\to 1}$ and $\boldsymbol{r}_1$ and feed them into a linear layer to obtain $\boldsymbol{a}_{2\to 1}$. Furthermore, the gate mechanism in Eq. (8) is used to determine the degree to which each element of $\boldsymbol{e}_{2\to 1}$ is accepted by $\boldsymbol{r}_1$. The $\boldsymbol{c}_{2\to 1}$ obtained from Eq. (8) is referred to as the **supplementary result** of $\boldsymbol{r}_2$ to $\boldsymbol{r}_1$. In this way, when $\boldsymbol{d}_{\mathrm{A}}$ and $\boldsymbol{d}_{\mathrm{V}}$ are used to replace $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ respectively, the supplement result of the video modality to the audio modality can be obtained, denoted as $\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}} \in \mathbb{R}^{D\times 1}$. Similarly, the supplement result of the audio modality to the video modality $\boldsymbol{c}_{\mathrm{A}\to\mathrm{V}} \in \mathbb{R}^{D\times 1}$ can also be obtained.

$$\boldsymbol{a}_{2\to 1} = \mathrm{Reshape}\left(\boldsymbol{W}_{2\to 1} \otimes \mathrm{Concat}\left(\boldsymbol{e}_{2\to 1}, \boldsymbol{r}_1\right)\right) \in \mathbb{R}^{D\times 1}, \tag{7}$$

where $\boldsymbol{W}_{2\to 1} \in \mathbb{R}^{D\times 2D}$ is a linear layer and $\mathrm{Concat}\left(\boldsymbol{e}_{2\to 1}, \boldsymbol{r}_1\right)$ means concatenating $\boldsymbol{e}_{2\to 1}$ and $\boldsymbol{r}_1$ along the row axis.

$$\begin{aligned} \boldsymbol{c}_{2\to 1} &= \uplus\left(\boldsymbol{r}_1, \boldsymbol{a}_{2\to 1}, \boldsymbol{e}_{2\to 1}\right) \\ &\triangleq \boldsymbol{r}_1 \oplus \mathrm{Sigmoid}\left(\boldsymbol{a}_{2\to 1}\right) \odot \boldsymbol{e}_{2\to 1} \in \mathbb{R}^{D\times 1}, \end{aligned} \tag{8}$$

where $\mathrm{Sigmoid}(\boldsymbol{a}_{2\to 1})$ operates along the row axis of $\boldsymbol{a}_{2\to 1}$.

Intuitively, after complementing each other, both of $\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}$ and $\boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}$ can be seen as the characterization of depressive cues. Therefore, the decision fusion strategy is adopted to predict BDI-II scores of the segments. In fact, Eq. (9) is used to obtain the estimation results of $\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}$ and $\boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}$ for depression levels. Furthermore, as shown in Eq. (10), we concatenate $\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}$ and $\boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}$, and feed them into a FC layer to obtain the weight coefficients for the decision fusion. Finally, the predicted value of BDI-II score for the segment is gained in the form of weighted sum, as shown in Eq. (11). And the average predicted score of segments is taken as the BDI-II score of the corresponding individual.

$$\begin{cases} s_{\mathrm{A}} = \boldsymbol{W}_{\mathrm{SA}} \otimes \boldsymbol{c}_{\mathrm{V}\to\mathrm{A}} \in \mathbb{R}^{1\times 1} \\ s_{\mathrm{V}} = \boldsymbol{W}_{\mathrm{SV}} \otimes \boldsymbol{c}_{\mathrm{A}\to\mathrm{V}} \in \mathbb{R}^{1\times 1} \end{cases}, \tag{9}$$

where $\boldsymbol{W}_{\mathrm{SA}} \in \mathbb{R}^{1\times D}$ and $\boldsymbol{W}_{\mathrm{SV}} \in \mathbb{R}^{1\times D}$ are two linear layers.

$$\boldsymbol{w} = \mathrm{Softmax}\left(\boldsymbol{W}_{\mathrm{f}} \otimes \mathrm{Concat}\left(\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}, \boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}\right)\right) \in \mathbb{R}^{2\times 1}, \tag{10}$$

where $\boldsymbol{W}_{\mathrm{f}} \in \mathbb{R}^{2\times 2D}$ is a FC layer and $\mathrm{Concat}\left(\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}, \boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}\right)$ is to concatenate $\boldsymbol{c}_{\mathrm{V}\to\mathrm{A}}$ and $\boldsymbol{c}_{\mathrm{A}\to\mathrm{V}}$ along the row axis.

$$s = \left[s_{\mathrm{A}}, s_{\mathrm{V}}\right] \otimes \boldsymbol{w} \in \mathbb{R}^{1\times 1}. \tag{11}$$

From the above content, one can observe that we explicitly stipulate the supplementary representation and propose the MC module based on that ***Stipulation***. This module jointly examines unimodal representation and supplementary representation, and adopts the gating mechanism to determine the acceptance degree of unimodal representation towards supplementary representation. Therefore, our proposed MC module can achieve effective modal complementarity and enhance the ability to characterize depression cues.

IV. Experiments

In this section, we first present the dataset used and metrics, then illustrate the effectiveness of our proposed modules through ablation experiments, and finally demonstrate the progressiveness of our method by comparing it with existing works.

A. Datasets

In the field of automatic depression level prediction, there are several commonly used datasets such as AVEC 2013, AVEC 2014, DAC-WOZ [27], Extended-DAIC [28], etc. However, only the AVEC 2013 and AVEC 2014 datasets provide raw video data. Therefore, this paper verifies the effectiveness and progressiveness of our proposed method on those two datasets because we need to use face images. Below, we briefly describe those two datasets.

For the AVEC 2013 dataset, it requires participants to sit in front of a computer and complete 14 interactive tasks shown on the screen. During the completion of tasks, the audio and video data of the participants are recorded for depression level prediction. In this way, the AVEC 2013 dataset collects 150 video samples containing audio from 82 participants. The duration of those samples ranges from 20 minutes to 50 minutes. When releasing the dataset, the author divide those 150 samples into three parts on average, namely the training set, development set, and testing set. It should be pointed out that all participants in the dataset speak German and are aged between 18 and 63. In addition, each participant is required to fill out a BDI-II scale before performing the interactive

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tasks, and the score on the scale is used as the label for the corresponding participant.

For the AVEC 2014 dataset, it is a part of the AVEC 2013 dataset and only selects audio and video data generated by participants under the Northwind and FreeForm tasks as samples. The duration of those samples ranges from 6 seconds to 4 minutes and 8 seconds. In those two tasks, each task has 150 samples. Similar to the AVEC 2013 dataset, the 150 samples are evenly divided into three parts by the publisher: training set, development set, and testing set. It should be noted that the audio and video data in these two tasks are derived from the same batch of participants, and the division structure regarding the training set, development set, and test set is exactly the same. Based on this, we combine the training set, development set, and testing set of those two tasks in the experiments. That is to say, in the following content, the AVEC 2014 training set, development set, and test set mentioned each contain 100 samples.

B. Metrics

When the AVEC 2013 and AVEC 2014 datasets are released, the authors use Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as shown in Eqs. (12) and (13) to measure the accuracy of depression level prediction. In subsequent works, researchers often use those two metrics to demonstrate the experimental performance of their methods. Hence, in this paper, we also adopt RMSE and MAE to measure the accuracy of our method for the task of predicting the level of depression. In addition, to fully demonstrate the effectiveness of our proposed module, R^2 and Symmetric Mean Absolute Percentage Error (SMAPE) as shown in Eqs. (14) and (15) are also used in ablation experiments.

$$\mathrm{RMSE} = \sqrt{\frac{1}{U}\sum_{u=1}^{U}\left(y_u - \widehat{y}_u\right)^2}, \tag{12}$$

where U refers to the number of samples. y_u and $\widehat{y}_u$ are the ground-truth and predicted BDI-II scores for the u-th sample, respectively.

$$\mathrm{MAE} = \frac{1}{U}\sum_{u=1}^{U}\left|y_u - \widehat{y}_u\right|. \tag{13}$$

$$R^2 = 1 - \frac{\sum_{u=1}^{U}\left(y_u - \widehat{y}_u\right)^2}{\sum_{u=1}^{U}\left(y_u - \overline{y}\right)^2}, \tag{14}$$

where $\overline{y}$ is the mean of ground-truth BDI-II scores of all samples.

$$\mathrm{SMAPE} = \frac{100\%}{U}\sum_{u=1}^{U}\frac{\left|y_u - \widehat{y}_u\right|}{\left(\left|y_u\right| + \left|\widehat{y}_u\right|\right)/2}. \tag{15}$$

C. Experimental Configuration

1) Preprocessing for Audio and Video Data: For the audio data, we separate the audio from the long-term video and resample it using a frequency of 16KHZ. Furthermore, the MFCC of the long-term audio is extracted by setting the parameters to (n_fft=512, hop_1ength=128, win_1ength=256, n_mfcc=39). In this way, the maximum value $f_{\max}$ and the minimum value $f_{\min}$ of each frequency band can be recorded. Next, we set a 3 second time window and a 50% repetition rate to segment the long-term video, separate the short-term audio and extract the corresponding MFCC. Finally, the normalized short-term MFCC is obtained through the max-min normalization process with $f_{\max}$ and $f_{\min}$ for each frequency band.

TABLE II
THE SETTING OF THE NETWORK LAYERS REGARDING OUR GEMC MODEL. "NF" AND "KS" REFER TO NUMBER OF FILTERS AND KERNEL SIZE, RESPECTIVELY. "(120)" REFERS TO THE NETWORK LAYER PARAMETERS SETTING FOR THE VIDEO MODALITY.

Module (Network)	Layers	NF (Neurons)	KS	Strides
GEMC network (Fig. 1)	Conv2D_1	1	[1, 9]	7
	Conv2D_2	1	[19, 1]	17
GE module & MHSA (Fig. 2)	Conv2D_3	1	[1, 1]	1
	Conv1D_1	39 (120)	1	1
	Conv1D_2	39 (120)	1	1
	Conv1D_3	39 (120)	1	1
	Conv2D_4	1	1	1
	Conv1D_4	1	1	1
MC module & Prediction (Fig. 3)	FC_1	64	-	-
	FC_2	64	-	-
	FC_3	64	-	-
	FC_4	1	-	-
	Conv1D_5	1	1	1
	FC_5	1	-	-

For the video data, we follow [22], [23], [29] to adjust the video sampling rate from 30fps to 6ps for producing good video analysis performance. And a 3-second window and 50% coverage are used to divide a long-term video into short-term video segments. Moreover, we adjust the resolution of each frame in the video segment from the original 640×480 to 480×480, and then transform it to 224×224, which is input into the pre-trained ResNet50 like in [21]–[23]. Next, the output of "avd_pool" layer is regarded as the encoding vector of a frame, so a video segment can be encoded into a sequence.

2) The Architecture for GEMC Network: In this paper, we predict the depression level using our constructed GEMC network, which mainly consists of two modules: GE and MC modules. Fig. 1 exhibits the architecture of GEMC network. Figs. 2 and 3 give the structure of GE and MC modules, respectively. Table II shows the parameter setting for different layers.

We take RMSE as the loss function to train our GEMC network. In this paper, we use the Keras deep learning framework to implement the training process. And the Adam [30] is selected as optimizer with default momentum values ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The weight decay and the initialization learning rate are set as 0.0001 and 0.00015, respectively.

D. Ablation Experiments

As mentioned earlier, we use the proposed GE and MC modules to construct a GEMC network for predicting individual depression levels. Therefore, in this section, some ablation experiments are performed to demonstrate the effectiveness of each component in the task of estimating depression severity.

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TABLE III
THE EXPERIMENTAL PERFORMANCE OF COMBINING MHSA WITH DIFFERENT MODULES ON AVEC 2013 AND AVEC 2014 DEVELOPMENT SETS. "GE" IS OUR PROPOSED GRADIENT EMBEDDING MODULE. "PE" IS SHORT FOR POSITION EMBEDDING. "CONV2D" REFERS TO REPLACING THE TWO GRADIENT CALCULATION TEMPLATES IN FIG. 2 WITH CONVOLUTIONAL LAYERS.

Models	Modalities	AVEC 2013				AVEC 2014			
		RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)	RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)
PE+MHSA	Audio	8.44	6.67	0.50	60.31%	8.14	6.42	0.53	54.71%
Conv2D+MHSA		8.24	6.58	0.52	59.43%	8.06	6.33	0.54	56.93%
GE+MHSA		7.30	5.90	0.62	60.41%	6.93	5.08	0.66	55.71%
PE+MHSA	Video	8.00	6.56	0.55	61.46%	7.68	5.70	0.59	52.12%
Conv2D+MHSA		7.77	5.26	0.57	**47.33%**	7.40	5.52	0.61	56.39%
GE+MHSA		6.93	5.77	0.66	62.90%	6.70	4.88	0.68	48.81%
PE+MHSA	Audio + Video	7.61	5.23	0.59	49.00%	7.44	5.19	0.61	55.51%
Conv2D+MHSA		7.18	5.31	0.64	51.91%	7.01	5.20	0.65	52.68%
GE+MHSA		**6.64**	**4.85**	**0.69**	48.83%	**6.49**	**4.52**	**0.70**	**44.45%**

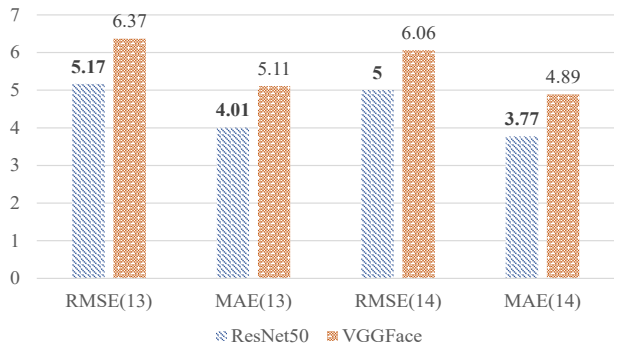

Fig. 4. Prediction accuracy of depression level using ResNet50 and VGGFace pre-trained models to encode facial images on AVEC 2013 and AVEC 2014 development sets. RMSE (13) and MAE (13) refer to the metrics of RMSE and MAE on AVEC 2013 development set. RMSE (14) and MAE (14) have the similar meaning.

1) The Effectiveness of ResNet50 for Encoding Facial Images: As shown in section III A, we use ResNet50 pre-trained with ImageNet to encode facial images and obtain the input sequence of video modality for our model. Perhaps one wonder why the VGGFace model is not used to encode facial images. To answer this question, we use those two pre-trained models to encode facial image sequences and input them into the GEMC network to predict BDI-II scores. Fig. 4 presents the corresponding experimental results.

Through comparison, we find that the experimental performance of ResNet50 is superior to VGGFace. This result is consistent with [12]. The main reason for this result is that VGGFace is good at extracting facial identity information, leading to similar representations extracted from different frames of the video for the same individual. This situation poses obstacles to examining the dynamic changes of the face. On the contrary, the ResNet50 extracts a representation of texture structure from the face, which helps predict BDI-II scores by examining facial dynamics.

2) The Effectiveness of GE Module: Considering the construction characteristics of the GE module, we clarify its effectiveness from the following three perspectives:

(1) From the perspective of position embedding substitution: As shown in Fig. 2, macroscopically speaking, the GE module can be seen as an alternative to PE. Hence, in this section, it is necessary to demonstrate the effectiveness of the GE module over PE in the automatic depression level prediction task. For this purpose, we conduct experiments on the development sets of AVEC 2013 and AVEC 2014, and the corresponding results are shown in Table III.

By observing those metric values, it is evident that the experimental performance of the GE module is superior to that of the PE. This is because the embedding process of the PE is the same for all sequences and does not have sequence adaptability, which poses some limitations in capturing differences among sequences of individuals with different levels of depression. In contrast, our proposed GE module embeds gradient information into the input sequence, which not only allows to perceive local changes of elements during long-term dependency modeling, but also adjusts the embedding matrix based on the input sequence. This is beneficial for capturing the discriminative dynamics presented by healthy and depressed individuals.

(2) From the perspective of convolution operation: From Fig. 2, one can see that we use a fixed template to perform convolution operations on the sequence to calculate the gradient of each element. Hence, we conduct experiments to demonstrate the effectiveness of this manner compared to convolutional layers. The corresponding results are shown in Table III.

By comparing "Conv2D+MHSA" and "GE+MHSA" in Table III, we can see that the experimental performance of the GE module is better than that of convolutional layers as a whole. This result also indicates that the parameters learned by convolutional kernels may not be able to better characterize local changes of elements and extract differences among individuals with different levels of depression than gradient templates. Interestingly, "Conv2D+MHSA" has better prediction accuracy than "PE+MHSA". On the one hand, this is due to the fact that convolutional layers can generate embedding results by examining the local changes in the input sequence itself, rather than using a fixed embedding matrix for all sequences like PE. On the other hand, the combination of convolutional layers and MHSA also enables the model to examine the input sequence from both local and global perspectives, which helps extract discriminative information among individuals with different levels of depression.

(3) From the perspective of the structure of the GE module and its combination with MHSA: From (1) and (2), it can be seen that the GE module is effective for the task of predicting depression levels. Next, we need to examine the rationality

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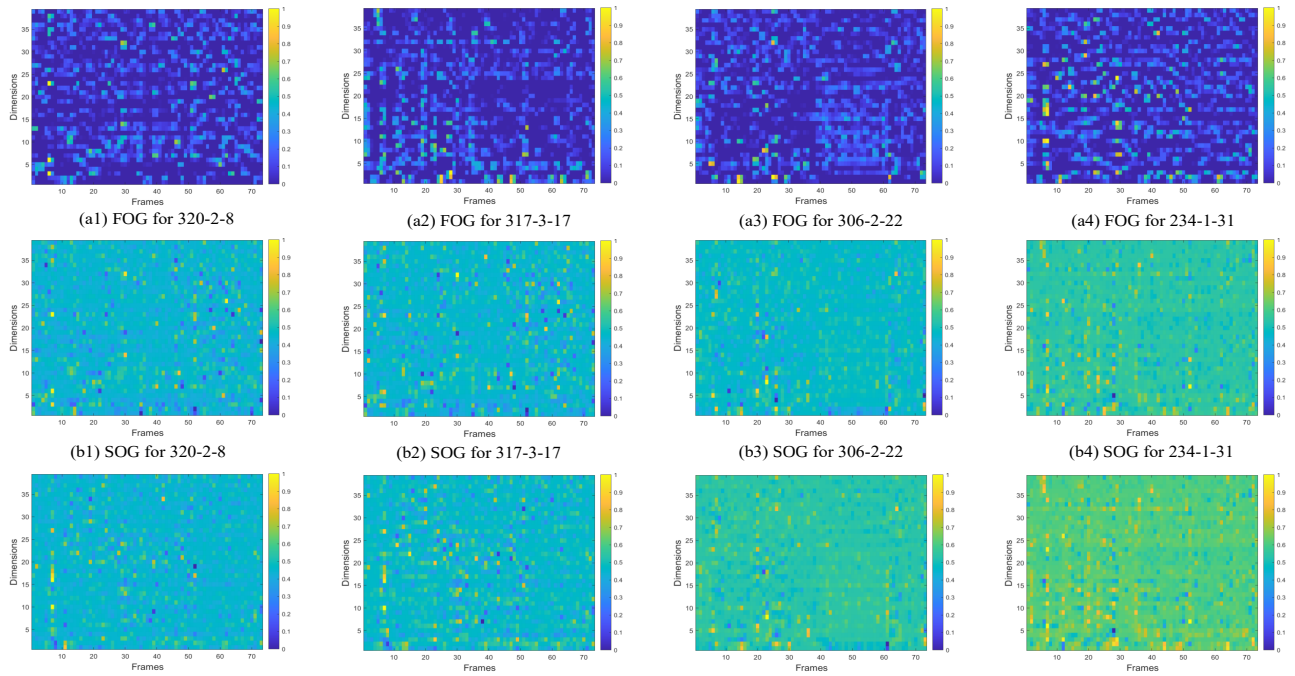
TABLE IV
EXPERIMENTAL PERFORMANCE OF DIFFERENT ORDER GRADIENT AND THEIR COMBINATION WITH MHSA ON AVEC 2013 AND AVEC 2014 DEVELOPMENT SETS. "FOG" AND "SOG" REFER TO THE FIRST-ORDER AND SECOND-ORDER GRADIENT. "GE" IS OUR PROPOSED GRADIENT EMBEDDING MODULE.

Models	Modalities	AVEC 2013				AVEC 2014			
		RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)	RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)
FOG	Audio	10.70	8.70	0.19	66.35%	10.49	8.54	0.23	71.03%
FOG+MHSA		8.48	6.46	0.49	59.11%	8.35	6.38	0.51	56.29%
SOG		10.56	8.74	0.21	66.59%	10.48	8.55	0.23	70.51%
SOG+MHSA		8.40	6.17	0.50	52.21%	8.31	6.11	0.51	51.64%
GE		9.64	7.95	0.34	67.92%	9.70	7.77	0.34	75.05%
GE+MHSA		7.30	5.90	0.62	60.41%	6.93	5.08	0.66	55.71%
FOG	Video	9.65	7.93	0.34	72.66%	9.40	7.88	0.38	66.51%
FOG+MHSA		7.34	6.11	0.62	57.73%	7.29	6.13	0.63	63.47%
SOG		9.47	7.83	0.37	72.76%	9.49	7.64	0.37	61.88%
SOG+MHSA		7.25	5.84	0.63	60.56%	7.11	5.63	0.64	56.16%
GE		9.10	7.54	0.42	66.50%	8.81	7.21	0.45	60.72%
GE+MHSA		6.93	5.77	0.66	62.90%	6.70	4.88	0.68	48.81%
FOG	Audio + Video	9.09	7.27	0.42	66.96%	9.07	7.15	0.42	63.14%
FOG+MHSA		7.00	5.07	0.65	56.17%	6.80	5.03	0.67	50.40%
SOG		9.05	7.25	0.42	62.11%	9.00	7.18	0.43	66.33%
SOG+MHSA		6.91	5.01	0.66	53.34%	6.77	5.04	0.68	48.93%
GE		8.89	7.04	0.44	67.53%	8.77	6.75	0.46	56.08%
GE+MHSA		**6.64**	**4.85**	**0.69**	**48.83%**	**6.49**	**4.52**	**0.70**	**44.45%**



(a1) FOG for 320-2-8 (a2) FOG for 317-3-17 (a3) FOG for 306-2-22 (a4) FOG for 234-1-31

(b1) SOG for 320-2-8 (b2) SOG for 317-3-17 (b3) SOG for 306-2-22 (b4) SOG for 234-1-31

(c1) GE for 320-2-8 (c2) GE for 317-3-17 (c3) GE for 306-2-22 (c4) GE for 234-1-31

Fig. 5. The gradient information is embedded into audio sequences of individuals with different depression levels. **"FOG" and "SOG"** refer to the use of first-order and second-order gradient to embed audio sequences. "GE" refers to our proposed gradient embedding module. "320-2-8" means that the BDI-II score of individual 320-2 is 8. "317-3-17", "306-2-22" and "234-1-31" have the same meaning.

of the structure of the GE module itself. That is to say, it is necessary to demonstrate the role of different order gradient for predicting depression levels. To this end, we conduct some ablation experiments and the results are presented in Table IV.

From Table IV, it can be seen that good prediction accuracy will not be achieved if only gradient operations, such as "1D", "2D", or "GE", are used. Whereas, the experimental performance of "GE" is better than "1D" and "2D", indicating that mixed order gradient is more effective for predicting BDI-II scores than using only first-order or second-order gradient. Additionally, one can observe that the combination of "1D" ("2D" or "GE") and MHSA, i.e. "1D+MHSA" ("2D+MHSA" or "GE+MHSA"), can rapidly reduce prediction errors compared to "1D" ("2D" or "GE"). This fact indicates that it is necessary to incorporate the MHSA module into the model for predicting depression levels. This comparison reflects that mixed order gradient has a better ability than first-order or second-order gradient to facilitate MHSA in extracting depression cues.

Furthermore, to more intuitively demonstrate the embedding effects of different methods on sequences, we provide a visual display in Fig. 5. Note that for the sake of simplicity, we only present the results of audio modality, and the results of video modality are similar. From the second and third rows of Fig.

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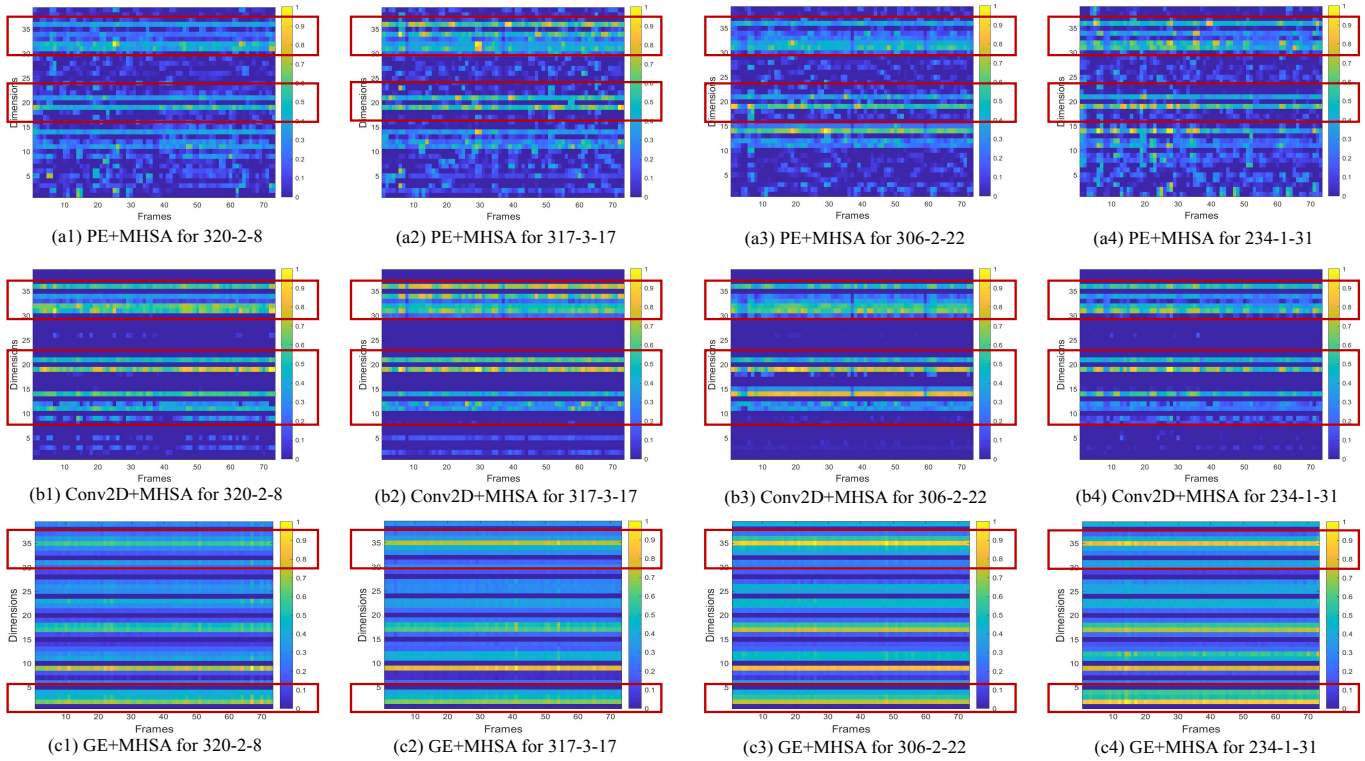

(a1) PE+MHSA for 320-2-8 (a2) PE+MHSA for 317-3-17 (a3) PE+MHSA for 306-2-22 (a4) PE+MHSA for 234-1-31

(b1) Conv2D+MHSA for 320-2-8 (b2) Conv2D+MHSA for 317-3-17 (b3) Conv2D+MHSA for 306-2-22 (b4) Conv2D+MHSA for 234-1-31

(c1) GE+MHSA for 320-2-8 (c2) GE+MHSA for 317-3-17 (c3) GE+MHSA for 306-2-22 (c4) GE+MHSA for 234-1-31

Fig. 6. The audio modal sequence is input into the models through the combination of different modules and MHSA, and the output results are visually displayed. The red boxes indicate the discriminative parts.

5, it can be seen that as the degree of depression deepens, the region of yellow elements also increases. Similar observations are difficult to reflect in the first row of Fig. 5. Moreover, it is easy to see that the discrimination of the third row of Fig. 5 is higher than that of the second row of Fig. 5. The reason for this situation is that although depression can lead to a lack of changes in an individual's speech, first-order gradient only has the advantage of capturing coarse-grained changes and is limited in extracting subtle changes. Therefore, the embedding effect of second-order gradient on elements is better than that of first-order gradient. In contrast, the GE module is able to integrate mixed order gradient and achieve a better embedding effect.

To demonstrate the effectiveness of combining GE and MHSA, we input the audio modal sequence into models combining different modules with MHSA, and visually display the output results of those architectures in Fig. 6. By comparing the three rows in Fig. 6, it is not difficult to find that "GE+MHSA" has the best discriminative performance. As shown in the parts circled by the red boxes in Fig. 6 (c1) - Fig. 6 (c4), the yellow line becomes increasingly prominent as the degree of depression deepens. Although similar phenomena are also reflected in Fig. 6 (a1) - Fig. 6 (a4) and Fig. 6 (b1) - Fig. 6 (b4), their discriminability is not as obvious as in Fig. 6 (c1) - Fig. 6 (c4). This fact indicates that our proposed GE module is more helpful for MHSA to perceive discriminative information between individuals with different levels of depression compared to PE and Conv2D layers. In other words, the gradient of elements is more effective in predicting depression levels than their position.

3) The Effectiveness of MC Module: In this paper, we provide the ***Stipulation*** of supplementary representation and construct the MC module to achieve complementarity between audio and video depression cue representations. Hence, it is necessary to demonstrate the effectiveness of the MC module through some ablation experiments.

(1) The rationality of modal supplementation: In this subsection, some ablation experiments are conducted to demonstrate the rationality of the "supplemental representation". Specifically, according to the ***Stipulation***, we examine the prediction accuracy of different modalities and supplemented depression cue representations. The corresponding experimental results are shown in Table V.

Compared with "A" and "V→A" ("V" and "A→V") in Table V, it can be seen that the prediction accuracy is improved. On the one hand, this indicates that the audio (video) modality indeed obtains effective supplementary information from the video (audio) modality. On the other hand, it also demonstrates the rationality of the stipulation of the supplementary representation we provide. Moreover, "A↔V " has the best experimental performance. This is because "A↔V " fuses the prediction results of "A→V" and "V→A", making the predicted BDI-II scores closer to the true values. In addition, we also achieve the representation fusion through the concatenation operation, and the corresponding metric values are shown in "[A, V]" in Table V. By comparing "[A, V]" and "A↔V", we can observe that the concatenation operation does not have better prediction results than the modal complementarity process we design. This can be explained as the concatenation operation being a linear combination among elements of two modal representa-

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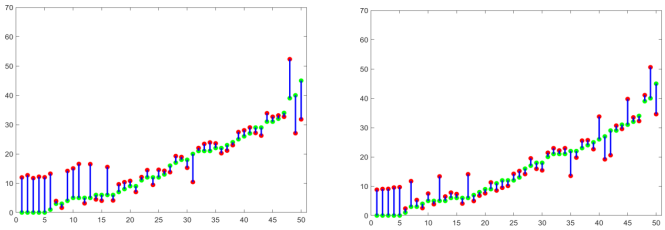
TABLE V
THE METRIC VALUES UNDER DIFFERENT MODAL RELATIONSHIPS ARE GIVEN FOR THE DEPRESSION LEVEL PREDICTION TASK ON THE AVEC 2013 AND AVEC 2014 DEVELOPMENT SETS. "A" AND "V" ARE SHORT FOR AUIDO AND VIDEO MODALITY. "V→A" ("A→V") REFERS TO THE SUPPLEMENT OF VIDEO MODALITY TO AUDIO MODALITY (AUDIO MODALITY TO VIDEO MODALITY). "A↔V" REFERS TO OUR PROPOSED MC MODULE. "[A, V]" REFERS TO CONCATENATING THE VECTORS OF AUDIO AND VIDEO MODALITIES INTO A LONG VECTOR.

Modal Relationships	AVEC 2013				AVEC 2014			
	RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)	RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)
A	7.30	5.90	0.62	60.14%	6.93	5.08	0.66	55.71%
V→A	7.02	5.68	0.65	55.25%	6.77	5.00	0.68	48.73%
V	6.93	5.77	0.66	62.90%	6.70	4.88	0.68	48.81%
A→V	6.88	5.53	0.67	55.03%	6.59	4.71	0.70	45.94%
A↔V	**5.17**	**4.01**	**0.81**	**44.06%**	**5.00**	**3.77**	**0.82**	**42.48%**
[A, V]	6.64	4.85	0.69	48.83%	6.49	4.52	0.70	44.45%

TABLE VI
THE IMPACT OF GATING MECHANISM ON PREDICTING DEPRESSION LEVELS IS EXAMINED ON THE DEVELOPMENT SETS OF AVEC 2013 AND AVEC 2014. THE UNDERLINED VALUES REPRESENT THE METRICS AFTER USING THE GATING MECHANISM. "✗" AND "✓" REFER TO NOT USING AND USING THE GATING MECHANISM, RESPECTIVELY. THAT IS TO SAY, "✗" MEANS REMOVING THE SIGMOID ACTIVATION FUNCTION IN EQ. (8).

Modal Relationships	Gating Mechanism	AVEC 2013				AVEC 2014			
		RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)	RMSE (↓)	MAE (↓)	R^2 (↑)	SMAPE (↓)
V→A	✗	7.24	5.82	0.63	57.89%	6.90	5.41	0.67	52.05%
	✓	7.02	5.68	0.65	55.25%	6.77	5.00	0.68	48.73%
A→V	✗	7.07	5.69	0.65	54.28%	6.64	5.07	0.69	53.66%
	✓	6.88	5.53	0.67	55.03%	6.59	4.71	0.70	45.94%
A↔V	✗	6.71	5.19	0.68	54.50%	6.87	4.91	0.67	48.55%
	✓	5.17	4.01	0.81	44.06%	5.00	3.77	0.82	42.48%



(a) The ground-truth v.s. predicted for [A, V]. (b) The ground-truth v.s. predicted for A ↔ V.

Fig. 7. The true values (green solid dots) and predicted values (red solid dots) under different modal supplementation scenarios in Table V on the AVEC 2013 development set. The blue vertical line indicates the difference between predicted and true values. The horizontal axis represents the subjects, and the vertical axis represents the BDI-II scores.

tions, which makes it difficult to clarify the complementary representation between the two modalities and to examine to what extent a modality needs to be supplemented.

Furthermore, in Fig. 7, we have plotted the predicted and ground-truth values for different modal supplementation scenarios in Table V. By observation, compared to "[A, V]" , one can see that "A↔V" is more able to make the predicted values closer to the ground-true values. However, it cannot be denied that the two modal supplementation manners do not perform well enough in predicting smaller BDI-II scores. This is related to the small number of subjects with smaller BDI-II scores in the dataset. Meanwhile, it is not an easy task for the neural network models to output exactly 0 after undergoing complex calculations.

(2) The role of gating mechanism: From Fig. 3, we use the gating mechanism to determine to what extent each element of the supplemental representation is accepted. Thus, in this subsection, the role of gating mechanism is examined. The corresponding results are presented in Table VI.

By comparing the metric values in Table VI, it is not difficult to find that the gating mechanism is effective in improving experimental performance. This fact indicates that the supplementation of audio (video) modality to video (audio) modality is not entirely useful for video (audio) modality. For example, for RMSE, "A" corresponds to a value of 7.30 in Table V. The corresponding values for "V→A (✗)" and "V→A (✓)" in Table VI are 7.24 and 7.02, respectively. The improvement in experimental performance is 0.82% and 3.84%, respectively. This result suggests that the gating mechanism can help extract useful and depression-related information from supplementary representations for modalities to be supplemented.

Furthermore, we present the results of gating under different modal supplementation scenarios in Fig. 8. From those results, it is not difficult for one person to see that overall, the supplementary demand obtained from video for audio is greater than that from audio for video. The reason is that the representations extracted from video modality have a stronger ability to characterize depression cues compared to the representations extracted from audio modality. At the same time, this fact also confirms that our proposed modal supplementation process, namely the MC module, is reasonable.

4) Significance Analysis about GE and MC Modules: To demonstrate the statistical effectiveness of our proposed GE and MC modules, we give the significance analysis results for those two modules in Table VII.

From Table VII, it can be seen that the p-values of our proposed GE and MC modules for those four metrics are all less than 0.05. In other words, the improvement in experimental performance of the GE and MC modules for the task of predicting depression levels is significant i.e., confidence levels exceeding 95%. Those results also indicate that the GE and MC modules are statistically effective. Furthermore, through careful comparison, it can be observed that the average confi-

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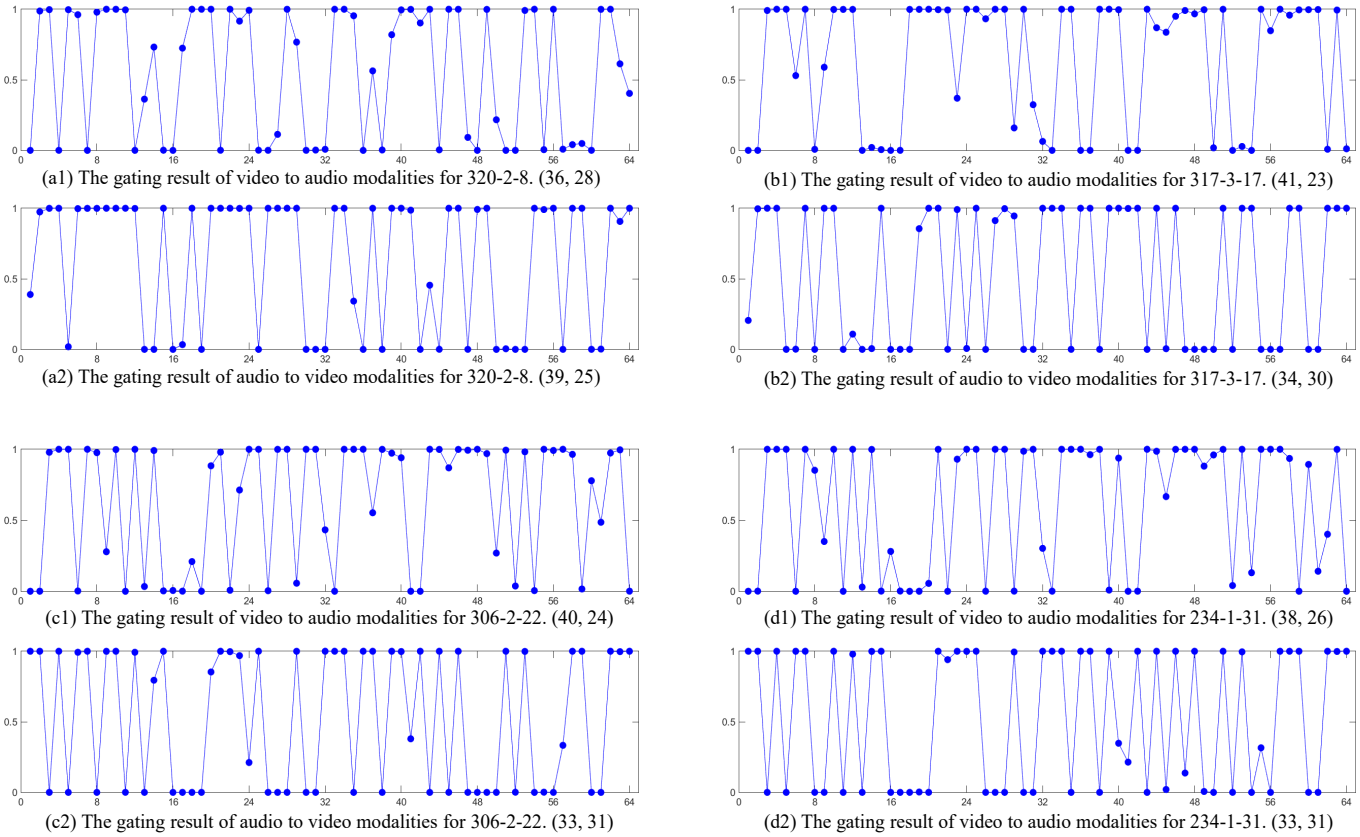

(a1) The gating result of video to audio modalities for 320-2-8. (36, 28)
(b1) The gating result of video to audio modalities for 317-3-17. (41, 23)
(a2) The gating result of audio to video modalities for 320-2-8. (39, 25)
(b2) The gating result of audio to video modalities for 317-3-17. (34, 30)
(c1) The gating result of video to audio modalities for 306-2-22. (40, 24)
(d1) The gating result of video to audio modalities for 234-1-31. (38, 26)
(c2) The gating result of audio to video modalities for 306-2-22. (33, 31)
(d2) The gating result of video to audio modalities for 234-1-31. (33, 31)

Fig. 8. The gating result of modal supplementation for individuals with different depression levels. "320-2-8" means that the BDI-II score of individual 320-2 is 8. "317-3-17", "306-2-22" and "234-1-31" have the same meaning. "(36, 28)" refers to 36 values greater than 0.5 and 28 values less than or equal to 0.5 in the gating results. "(39, 25)", "(41, 23)", "(34, 30)", "(40, 24)", "(33, 31)", "(38, 26)" and "(33, 31)" have the same meaning.

TABLE VII
THE SIGNIFICANCE ANALYSIS RESULTS OF THE PROPOSED GE AND MC MODULES ON THE METRICS.

Modules	Metrics	p-values	Confidence level
GE module	RMSE	0.0378	96.22%
	MAE	0.0378	96.22%
	R^2	0.0368	96.32%
	SMAPE	0.0421	95.79%
MC module	RMSE	0.0443	95.57%
	MAE	0.0490	95.10%
	R^2	0.0480	95.20%
	SMAPE	0.0424	95.76%

dence level of the GE module for the four metrics is greater than that of MC module. This further indicates that it is more important to consider local changes of elements in modeling long-range dependencies than modal complementarity.

5) Complexity Analysis: In this subsection, we analyze the complexity of different models. The corresponding results are shown in Table VIII. It should be noted that in the field of depression level prediction, there are few works to demonstrate their model parameters and computational complexity. Hence, we only present the complexity of the models involved in the ablation experiments.

As shown in Table VIII, "PE+MHSA+[A, V]" has the least number of parameters and the lowest computational complexity. This is because PE itself does not require training parameters and only needs to perform addition with the same number of elements as the input matrix. Meanwhile, [A, V] requires fewer parameters and calculations compared to the

TABLE VIII
PARAMETERS AND COMPUTATIONAL COMPLEXITY OF DIFFERENT MODELS. "A ↔ V" REFERS TO OUR PROPOSED MC MODULE. "[A, V]" MEANS TO THE CONCATENATION OF AUDIO AND VIDEO MODALITIES.

Models	Parameter	FLOPs
PE+MHSA+[A, V]	51133	7369508
Conv2D+MHSA+[A, V]	51142	7454627
GE+MHSA+[A, V]	51136	7429592
GE+MHSA+A↔V	77732	7482970

MC module. In addition, since Conv2D and GE module only have parameters in the convolutional layer, the number of parameters has not increased significantly. However, the element-wise convolution brings a significant increase in computational complexity. Due to the involvement of multiple linear layers in the MC module, "GE+MHSA+A↔V" requires the highest number of parameters and calculations. However, it is interesting that the large number of parameters do not lead to overfitting of "GE+MHSA+A↔V" and affect the experimental performance, which also illustrates the effectiveness of our method.

6) Comparison and Discussion with Existing Methods: In order to illustrate the progressiveness of our method in predicting depression levels, we conduct experiments on AVEC 2013 and AVEC 2014 test sets. The corresponding results are presented in Tables IX and X. By comparison, in the mass, our method has the best experimental performance.

From the perspective of model architecture: It cannot be denied that our GEMC network can be seen as an extension

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TABLE IX
EXPERIMENTAL PERFORMANCE OF OUR METHOD AND EXISTING WORKS ON THE AVEC 2013 TEST SET. BOLDED NUMBERS ARE THE BEST RESULTS.

Modality	Methods	Years	RMSE	MAE
Audio	Baseline [14]	2013	14.12	10.35
	DCNN [31]	2018	10.00	8.20
	TFCAV [32]	2021	8.32	6.26
	TDCAN [33]	2021	9.22	6.90
	CSE [34]	2022	9.28	6.79
	Hybrid Network [35]	2022	9.63	7.51
	STN [9]	2023	8.16	6.70
	FTAN [36]	2023	9.58	7.42
	STA [10]	2023	9.50	7.14
	WavDepressionNet [37]	2024	8.20	6.14
	Time-Domain [38]	2025	9.24	8.26
	SpectrumFormer [39]	2025	8.12	6.09
	SIMMA [40]	2025	8.67	6.69
	DAMC Network (Ours)	-	8.26	6.18
Video	Baseline [14]	2013	13.61	10.88
	Spectral [41]	2022	8.10	6.16
	SE-TOV [42]	2022	7.42	6.09
	LQGDNet [43]	2023	8.20	6.38
	DAER [12]	2023	8.13	6.28
	AVA-DepressNet [44]	2023	7.26	5.97
	MDN [45]	2023	7.55	6.24
	STN [46]	2023	7.32	5.90
	STA-DRN [47]	2024	7.98	6.15
	DMSN [48]	2024	7.66	6.14
	DepressionMLP [49]	2024	7.49	5.43
	MTDAN [50]	2024	8.08	6.14
	Graph Representation [51]	2025	7.57	5.95
	SpatioTemporal Routing [52]	2025	6.74	5.38
	MemRank [53]	2025	7.78	5.82
	LMTformer [5]	2025	7.75	6.12
	SIMMA [40]	2025	7.97	5.94
	DepMGNN [54]	2025	7.08	5.41
	DAMC Network (Ours)	-	7.51	5.82
Audio & Video	A-V System [55]	2014	11.19	9.09
	GMM-UBM [56]	2013	10.62	/
	MHH+PLS [57]	2013	10.96	8.72
	Ensemble CCA [58]	2014	9.44	7.68
	Two-stage [59]	2017	8.29	6.75
	STA [10]	2023	8.16	6.14
	AVA-DepressNet [44]	2023	7.99	6.23
	CMAFN [60]	2023	6.60	5.26
	STN [9]	2023	6.83	5.38
	Transformer-based [4]	2024	9.00	6.60
	Graph Representation [61]	2024	8.31	6.80
	Interpretable [62]	2025	7.89	6.15
	Video-attention [63]	2025	8.06	6.26
	STE-Mamba [64]	2025	6.94	5.08
	SIMMA [40]	2025	6.97	5.35
	MFMamba [65]	2025	7.05	6.11
	MMDepNet [66]	2025	8.11	6.14
	DAMC Network (Ours)	-	**6.42**	**5.06**

TABLE X
EXPERIMENTAL PERFORMANCE OF OUR METHOD AND EXISTING WORKS ON THE AVEC 2014 TEST SET. BOLDED NUMBERS ARE THE BEST RESULTS.

Modality	Methods	Years	RMSE	MAE
Audio	Baseline [15]	2014	12.56	10.03
	DCNN [31]	2018	9.99	8.19
	TFCAV [32]	2021	9.25	7.49
	TDCAN [33]	2021	8.90	7.08
	CSE [34]	2022	9.61	7.13
	Hybrid Network [35]	2022	9.40	7.37
	STN [46]	2023	8.46	6.95
	FTAN [36]	2023	9.60	7.31
	STA [10]	2023	9.13	7.65
	WavDepressionNet [37]	2024	8.61	6.60
	Time-Domain [38]	2025	9.20	8.39
	SpectrumFormer [39]	2025	8.31	6.06
	SIMMA [40]	2025	9.55	7.31
	DAMC Network (Ours)	-	8.08	6.02
Video	Baseline [15]	2014	10.86	8.86
	Spectral [41]	2022	7.15	6.95
	SE-TOV [42]	2022	7.39	5.87
	LQGDNet [43]	2023	7.84	6.08
	DAER [12]	2023	8.07	6.14
	AVA-DepressNet [44]	2023	7.30	5.99
	MDN [45]	2023	7.65	6.06
	STN [9]	2023	6.98	5.75
	STA-DRN [47]	2024	7.75	6.00
	DMSN [48]	2024	7.50	5.69
	DepressionMLP [49]	2024	7.27	5.63
	MTDAN [50]	2024	7.93	6.35
	Graph [51]	2025	7.18	5.86
	SpatioTemporal Routing [52]	2025	6.83	5.09
	MemRank [53]	2025	7.69	5.77
	LMTformer [5]	2025	9.97	6.05
	SIMMA [40]	2025	7.31	5.35
	DepMGNN [54]	2025	6.28	4.99
	GEMC Network (Ours)	-	7.11	5.23
Audio & Video	Baseline [15]	2014	9.89	7.89
	Ensemble CCA [58]	2014	9.61	7.69
	SVR [67]	2014	8.99	/
	GMM+ELM [68]	2014	8.12	6.31
	m-BAM [69]	2019	7.47	5.78
	STA [10]	2023	7.03	5.21
	AVA-DepressNet [44]	2023	6.83	5.32
	CMAFN [60]	2023	**6.04**	5.16
	STN [9]	2023	6.16	5.03
	Transformer-based [4]	2024	9.45	7.05
	Graph Representation [61]	2024	8.18	6.59
	Interpretable [62]	2025	8.11	6.14
	Video-attention [63]	2025	7.73	5.95
	STE-Mamba [64]	2025	6.77	5.10
	SIMMA [40]	2025	6.95	5.14
	MFMamba [65]	2025	6.71	5.16
	MMDepNet [66]	2025	7.89	6.15
	GEMC Network (Ours)	-	6.23	**5.00**

of the Transformer structure. Therefore, the good experimental performance indicates that the structure of the Transformer is suitable to the task of predicting individual depression levels. It should be pointed out that the experimental accuracy of our GEMC network is better than that of Transformer-based [4] and LMTformer [5]. This is because those two works do not take into account the dynamic information embedding in the process of long-range modeling.

From the perspective of local-global feature extraction: CMAFN [60], STE-Mamba [64], and MFMamba [65] exhibit relative advantages in terms of metrics. This indicates that the combination of local and global features promotes the models to capture depression cues. However, the experimental performance of those works is not consistently better than our method. This is because our method embeds the gradient information into the input sequence to enable the model to perceive the local dynamic while examining the global, rather than just alternating between local and global information extraction processes.

From the perspective of multimodal complementarity: Most of the multimodal methods listed in the above tables achieve modal complementarity through the feature concatenation or attention mechanism. Although those methods have achieved good results, their experimental performance is weaker than our GEMC network. This result suggests that the stipulation of supplementary representation we provide is reasonable and it is necessary to consider to what extent the supplementary representation is accepted.

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V. Conclusion

The research results of physiology and psychology indicate that individuals with different levels of depression exhibit discriminative temporal patterns of audio and video modalities. To this end, we construct a GEMC network to predict an individual's depression level. In our constructed GEMC network, we propose GE and MC modules. The GE module embeds gradient information into the sequence to enable MHSA to perceive temporal changes of elements while the examination of long-range dependencies, which enhances the model's ability to capture speech and facial dynamics of individuals with different depression levels. The MC module, based on the supplementary representation stipulation we provide, enables audio or video modalities to automatically determine the degree of acceptance of supplementary representations to achieve effective utilization of supplementary representations, thereby enhancing the ability to characterize depressive cues. On the AVEC 2013 and AVEC 2014 dataset, we obtain the metrics of RMSE=6.42/6.23 and MSE=5.06/5.00, which verify the effectiveness and progressiveness of our method. However, the facial images we use are prone to privacy risks, which makes it difficult for our model to be accepted by users and reduces its practical value. Therefore, in the future, it is necessary to design a model with high accuracy and less likely to expose user identities.

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